Tanmay Bhuskute

+1 (732) 858-3027 | tanmay.v.bhuskute@gmail.com | github.com/Tanmayb05 | linkedin.com/in/tanmay-bhuskute
Sunnyvale, CA

SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production backend engineering experience building distributed systems, REST APIs, and cloud platforms using Python, Java, SQL, AWS, and Kubernetes. Skilled in large-scale system design, production debugging, and DevOps practices.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, C/C++, SQL, MySQL, Go, JavaScript, TypeScript.

Backend & APIs: REST API design and integration, Object-Oriented Design, Data Structures, Algorithms, Complexity Analysis, Microservices, PostgreSQL, MySQL.

Developer Tools: Git, GitLab CI/CD, Jenkins, AWS, Docker, Kubernetes, ArgoCD, Helm, Terraform, Linux, Bash, SonarQube, Azure, Google Cloud Platform.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across large-scale distributed AWS environments.
- Executed **44 production environment upgrades** in 4 months, debugging and monitoring live infrastructure across Linux, EC2, databases, and Teamcenter X services to resolve production reliability issues.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%** through scalable, resilient rollout architecture.
- Migrated **40+ production environments** from process-based deployments to Kubernetes using ArgoCD, Helm, Docker, GitLab CI/CD, Amazon EFS, and HashiCorp Vault.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; collaborated across Engineering teams and mentored **5 software engineers** through structured onboarding.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern —- Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [PySpark, Delta Lake, scikit-learn, FastAPI, SQL] Jul 2026

- Built large-scale distributed ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** dataset); designed two-stage model improving MAPE **18.77% to 16.52%** and cutting RMSE **43%**.
- Built RESTful API (Python, FastAPI) serving real-time predictions, integrated with PostgreSQL and Redis caching, containerized with Docker.

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG pipeline** over Google Gemini 2.5-flash, forcing citation-backed answers to eliminate hallucination in high-stakes decisions.
- Designed **17-tool MCP server** exposing structured API tool-calling for PA extraction, patient-readiness evaluation, and cross-payer comparison.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).